💡 Executive Summary: Telco Customer Churn Analysis

1. Project Objective

The goal of this project was to analyze customer behavior within a telecommunications company to identify the key factors leading to **Churn** (customers leaving the service). By understanding these drivers, the business can implement targeted retention strategies.

2. Data Overview

- **Dataset Size:** 7,043 customers with 21 different features.
- **Key Variables:** Customer demographics (gender, senior citizen), account information (tenure, contract, payment method), and subscribed services (Internet, Tech Support, Streaming).
- **Target Variable:** Churn (Yes/No).

3. Key Findings (Insights)

- **Contract Sensitivity:** Customers on **Month-to-Month** contracts have a significantly higher churn rate compared to those on one- or two-year contracts.
- **Service Impact:** Users without **TechSupport** and **OnlineSecurity** are more likely to leave, suggesting these "peace of mind" services are high value for retention.
- **Financial Drivers:** There is a clear correlation between high **MonthlyCharges** and churn. Customers paying higher amounts are more sensitive to leaving, especially if they are on fiber-optic internet.
- **Tenure:** New customers (tenure < 6 months) are at the highest risk. If a customer stays past the first year, their loyalty increases substantially.

4. Technical Skills Demonstrated

- **Data Cleaning:** Handled data types (converting TotalCharges from object to numeric) and managing missing values.
- **Univariate Analysis:** Explored the distribution of individual services using organized subplots.
- **Bivariate & Multivariate Analysis:** Used Boxplots, KDE plots, and Heatmaps to visualize the relationship between billing, tenure, and churn.
- **Data Visualization:** Utilized Matplotlib and Seaborn to create professional, readable charts with clear labeling and aesthetic styling.

5. Business Recommendations

- **Incentivize Long-term Contracts:** Offer discounts to move month-to-month customers into at least a 1-year agreement.
- **Bundle Security Services:** Provide "Security & Support" bundles for free or at a discount for high-risk new users to increase the "stickiness" of the service.
- **High-Bill Watchlist:** Monitor customers with high monthly charges and low tenure for proactive outreach or loyalty rewards.